

SRMVEC
DEPARTMENT OF AIDS
Semester End Examination
(Regulations 2023)
AD3445 – AI1

Branch: B.Tech | Date: 01.02.2026 | Max. Marks: 100 | Duration: 3 Hrs

Answer ALL Questions
(100 Marks)

PART A — (10 × 2 = 20 marks)

1. Define a Rational Agent and list the components of a PEAS description. [CO1] [BTL1-Remember] (2)
2. Differentiate between Breadth-First Search (BFS) and Depth-First Search (DFS) in terms of completeness and optimality. [CO1] [BTL2-Understand] (2)
3. What are the requirements for a heuristic function to be admissible in A* search? [CO2] [BTL2-Understand] (2)
4. Explain the concept of 'Pruning' in the context of game tree search. [CO2] [BTL2-Understand] (2)
5. Represent the following sentence in First Order Predicate Logic (FOPL): 'Every student who takes AI passes the exam'. [CO3] [BTL3-Apply] (2)
6. Define 'Unification' in the context of inference in logic. [CO3] [BTL1-Remember] (2)
7. State the Bayes' Rule formula and identify each term. [CO4] [BTL1-Remember] (2)
8. What is a Conditional Probability Table (CPT) in a Bayesian Network? [CO4] [BTL2-Understand] (2)
9. Distinguish between Supervised and Unsupervised Learning with one example each. [CO5] [BTL2-Understand] (2)
10. What is the role of an 'Activation Function' in an Artificial Neural Network? [CO5] [BTL2-Understand] (2)

PART B — (5 × 16 = 80 marks)

11(a) (i) Explain the various types of Agent Programs with neat diagrams. (ii) Formulate the 8-Queens problem as a state-space search problem. Define the states, initial state, actions, transition model, and goal test.

OR

(b) Explain the Uniform Cost Search (UCS) algorithm. Illustrate its working on a graph of your choice and discuss its time and space complexity.

12(a) Explain the A* Search algorithm in detail. Prove that A* is optimal if the heuristic function $h(n)$ is admissible.

OR

(b) (i) Describe the Minimax algorithm for game playing with a suitable game tree. (ii) Apply Alpha-Beta pruning on a game tree of depth 3 with branching factor 2 and show which nodes are pruned.

13(a) (i) Explain the process of converting a FOPL sentence into Conjunctive Normal Form (CNF) with an example. (ii) Describe the Forward Chaining and Backward Chaining inference techniques in Propositional Logic.

OR

(b) Discuss the different ways of representing knowledge in AI, specifically focusing on Semantic Networks and Frames. Provide examples for each.

14(a) Construct a Bayesian Network for the 'Burglary' problem (Burglary, Earthquake, Alarm, JohnCalls, MaryCalls). Explain how the joint probability distribution is calculated and perform a sample inference calculation.

OR

(b) (i) Explain the components and working of a Hidden Markov Model (HMM). (ii) Describe the concept of

Dempster-Shafer Theory for handling uncertainty.

15(a) Explain the ID3 Decision Tree induction algorithm. Use the concept of Information Gain and Entropy to illustrate how the root node is selected for a sample dataset.

OR

(b) (i) Explain the architecture of a Multi-Layer Perceptron (MLP) and the Backpropagation learning rule. (ii) Briefly discuss the Reinforcement Learning framework (Agent-Environment interaction).

Faculty In-Charge

Verified by

HOD